AUTOMATION SYSTEM DESIGN

AUTOMATED PACKAGING SYSTEM

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Fig: Automated Packaging System

1.How robotics works in this system

1. Object detection

An IR sensor detects when a product arrives on the conveyor belt and sends a signal to the controller.

2. Product movement

A motor moves the product to the required position. A robotic arm can also pick up and place products.

3. Measurement and checking

A load cell measures the product's weight. The controller checks whether it meets the programmed requirements.

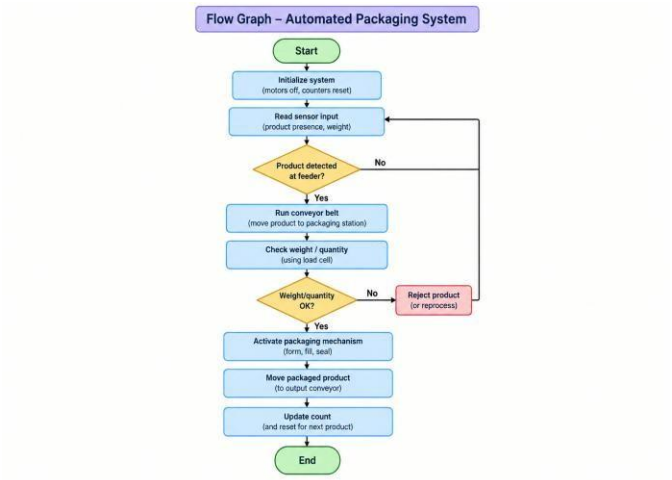
4. Packaging and sealing

A packaging mechanism wraps the product, and a heat-sealing mechanism seals the package. These functions may be automated without a robotic arm.

2. Main robotics concepts involved

Concept	Role in the system
Sensors	Detect products and measure weight
Control system	Arduino or PLC makes decisions
Actuators	Motors and servo motors perform movement
Automation	Executes packaging steps automatically
Feedback control	Uses sensor readings to adjust operation
Robotic manipulation	A robotic arm can pick, place, or sort products

Here is the simple flow graph:



3. What is the role of robotics in this system?

Robotics combines mechanical engineering, electronics, sensors, and programming to perform tasks automatically.

- Mechanical system: The robotic arm and joints provide movement.
- Electrical system: Motors and actuators drive the arm.
- Sensors: Detect boxes and provide position information.
- Control system: Coordinates the robot's movements.
- End effector: The suction-cup gripper picks up the boxes.
- Automation: The robot repeats programmed tasks with minimal manual intervention.

4. Advantages

- Increases packaging speed and productivity.
- Reduces repetitive manual work.
- Improves placement accuracy and consistency.
- Can operate continuously with appropriate supervision and maintenance.
- Reduces direct handling of packages. **5. Real-world applications**

This technology is used in:

- E-commerce warehouses for handling parcels.
- Food and beverage industries for packing products.